



Matteo Neri

PhD student

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Summary

I am a mathematician and complex systems physicist pursuing a PhD at the intersection of Complex Systems Physics, Computational Neuroscience, and AI. Working between Aix-Marseille University and Imperial College London, I study the emergence, structure, and limits of intelligence in artificial and biological systems using information theory and graph theory as unifying frameworks.

My research focuses on understanding information integration, causal structure, and representational properties in complex systems, as well as mechanistic interpretability, recently investigating superposition through an information-theoretic lens. I also contribute to open-source code, co-developing two Python packages (published in JOSS) that make advanced information- and graph-theoretic methods accessible for analysing and evaluating intelligent systems. I enjoy conducting research and communicating it through scientific writing. In the first two years of my PhD, I co-authored 12 papers, including publications in *Nature Communications*, *Physical Review Letters*, and the *European Journal of Neuroscience*.

Education

- *PhD candidate - visiting*, Machine Intelligence and Neural Dynamics (MIND) lab, Department of Computing, **Imperial College London**, London, UK
Aug 2025 – Present
Nov 2024 – Dec 2024
- *PhD Candidate*, **Institut de Neurosciences de la Timone**, Marseille, France,
Oct 2023 – Present
- **MSc. in Physics of Complex Systems Physics (with honors)**, University of Turin,
Jan 2021 – Apr 2023

- **BSc. in Mathematics**, University of Bologna (UNIBO), Italy,
Nov 2017 – Oct 2020
- **Erasmus Programs**, Aix-Marseille Université (AMU), France, Sep 2021 – Jul 2022;
Universitat de Barcelona (UB), Spain,
Sep 2019 – July 2020

Research Experience

- *Research Assistant*, **Institut de Neurosciences de la Timone**, Marseille, France,
Apr 2023 – Nov 2023
- *Intern Student*, **Institut de Neurosciences de la Timone**, Marseille, France,
Nov 2022 – Apr 2023
- *Intern Student*, **CENTAI**, Turin, Italy,
Oct 2022 – Apr 2023
- *Intern Student*, **Institut de Neurosciences des Systèmes**, Marseille, France,
Mar 2022 – Sep 2022

Teaching and Scientific Communication

- **Teaching Assistant**, Aix-Marseille Université, Marseille (France),
Sep 2023 - Jun 2024
 - L3S5 - HPS5U07 - Méthodes et statistiques (36h)
 - L3S5 - HPS5U07 – Initiation à la psychologie Mathématique (30h)
- **Math Tutor**, Associazione diamo i numeri (~120 hours in total), Bergamo (Italy),
2021–Present
- **The AI for the people podcast** – online,
Sep 2021 - Sep 2022
I participated in the organization of a podcast. The main topics discussed relate to **AI safety** and the value of AI for the future of humanity.

Grants and Awards

- *BIFONDS travel grant* (2025) — Funding for 3 months exchange, **~€3000**
- *Neuroscool PhD funding* (2023) — PhD scholarship, **~€100K**
- *Interdisciplinary College IK* (2023) — Full stipend recipient, **~€400**
- *Poster award at NeuroDays* (2023) — prize of **€150**

Open-source software (peer-reviewed in JOSS)

Matteo Neri, Dishie Vinchhi, Christian Ferreyra, Thomas Robiglio, Onur Ates, Marlis Ontivero-Ortega, Andrea Brovelli, Daniele Marinazzo, and Etienne Combrisson (2024) *HOI: A Python toolbox for high-performance estimation of Higher-Order Interactions from multivariate data*. *Journal of Open Source Software (JOSS)*

Article DOI: <https://doi.org/10.21105/joss.07360> Github: [brainets/hoi: Higher-Order Interactions](https://github.com/brainets/hoi)

Maxime Lucas, A Townsend-Teague, **Matteo Neri**, Simone Poetto, A Morris, B Habermann (2023) *Phasik: a Python package to identify system states in partially temporal networks*. *Journal of Open Source Software*

Article DOI: <https://doi.org/10.21105/joss.05872> Gitlab: [habermann_lab / phasik · GitLab](https://gitlab.com/habermann_lab/phasik)

List of peer-reviewed publications

ORCID: 0009-0007-0998-552X

Thomas Robiglio, **Matteo Neri**, Davide Coppes, Cosimo Agostinelli, Federico Battiston, Maxime Lucas, Giovanni Petri (2024) *Synergistic signatures of group mechanisms in higher-order systems* *Physical Review Letters*

Article DOI: <https://doi.org/10.1103/PhysRevLett.134.137401>

Etienne Combrisson, Ruggero Basanisi, **Matteo Neri**, Guillaume Auzias, Giovanni Petri, Daniele Marinazzo, Stefano Panzeri, Andrea Brovelli (2024) *Higher-order and distributed synergistic functional interactions encode information gain in goal-directed learning*. *Nature Communications*

Preprint DOI: <https://doi.org/10.1101/2024.09.23.614484>

Matteo Neri, Andrea Brovelli, Samy Castro, Fausto Fraiso, Marilyn Gatica, Ruben Herzog, Pedro A.M. Mediano, Ivan Mindlin, Giovanni Petri, Daniel Bor, Fernando E. Rosas, Antonella Tramacere, and Mar Estarellas (2024) *A taxonomy of neuroscientific strategies based on interaction orders*. *European Journal of Neuroscience*

Article DOI: <https://doi.org/10.1111/ejn.16676>

Claudio Runfola, **Matteo Neri**, Daniele Schön, Benjamin Morillon, Agnès Trébuchon, Giovanni Rabuffo, Pierpaolo Sorrentino, Viktor Jirsa (2024) *Complexity in speech and music listening via neural manifold flows* **Network Neuroscience**

Article DOI: https://doi.org/10.1162/netn_a_00422

Hasnae Agouram, **Matteo Neri**, Marianna Angiolelli, Damien Depannemaecker, Jyotika Bahuguna, Antoine Schwey, Jean Régis, Romain Carron, Alexandre Eusebio, Nicole Malfait, Emmanuel Daucé, and Pierpaolo Sorrentino (2024) *L-Dopa induced changes in aperiodic bursts dynamics relate to individual clinical improvement in Parkinson's disease* **npj Parkinson's Disease**

Preprint DOI: <https://doi.org/10.1101/2024.06.14.24308683>

Carlos Coronel-Oliveros, Marilyn Gatica, Rubén Herzog, **Matteo Neri** (2025) *An integrated computational framework for diversity-sensitive personalized medicine* **IBRO Neuroscience**

Preprint DOI: <https://doi.org/10.1016/j.neuroscience.2025.08.019>

Review & Resubmit (R&R) major or minor revision in process

Andrea Santoro, **Matteo Neri**, Simone Poetto, Davide Orsenigo, Matteo Diano, Marilyn Gatica, Giovanni Petri (2025) *Beyond Pairwise Interactions: Charting Higher-Order Models of Brain Function* **R&R Nature Communications**

Preprint DOI: <https://doi.org/10.1101/2025.06.24.661306>

Matteo Neri, Claudio Runfola, Noemie AG te Rietmolen, Pierpaolo Sorrentino, Daniele Schon, Benjamin Morillon, Giovanni Rabuffo (2023) *Neuronal Avalanches in Naturalistic Speech and Music Listening* **R&R Imaging Neuroscience**

Preprint DOI: <https://doi.org/10.1101/2023.12.15.571888>

Camille Mazzara, Giuditta Gambino, Gian Marco Duma, **Matteo Neri**, Giuseppe Giglia, Tommaso Piccoli, Fabrizio Guajana, Albert Comelli, Antonino Tuttolomondo, Michele Migliore, Pierpaolo Sorrentino (2025) *Brain Fluidity as a Functional Marker of Tau-Related Neurodegeneration in Alzheimer's Disease*

Preprint DOI: <https://doi.org/10.1101/2025.07.04.25330744>

Yuri Gardinazzi, Roger González March, Suprabhath Kalahasti, Andrea Montaña Ramirez, **Matteo Neri**, Cicely Nguyen, Giovanni Palermo, Erik Weis, Katharina Ledebur, Elma Dervić (2025) *Characterization of diseases in temporal comorbidity networks*

Preprint DOI: <https://arxiv.org/html/2506.22136v1>

Talks and presentations

- **Tutorial for Beyond the Edge Workshop**
Marseille, France (October 2025)
I presented a tutorial (30min) about our toolbox HOI (<https://brainets.github.io/hoi/>).
- **Conference on Complex Systems**
Siena, Italy (September 2025)
The biggest conference on complex systems.
My contribution: **Regular talk and satellite talk**
- **Complexity72h**
Madrid, Spain (June 2025)
I presented a tutorial (30min) about our recently published Python toolbox (<https://brainets.github.io/hoi/>).
- **HOI-BrainMod**
Institute for Advanced Studies (IAS), *Amsterdam, Netherlands (May 2025)*
Workshop on Brain Modeling and Higher-Order Interactions
(<https://ias.uva.nl/content/events/2025/05/hoi-brainmod-workshop-on-brain-modeling-and-high-order-interactions.html>).
My contribution: **Organizer, talk, and a tutorial.**
- **Interdisciplinary College (IK) 2024**
Interdisciplinary forum that combines education and research in fields like AI, neuroscience, and cognitive science.
My contribution: **Talk and tutorial** (~1.5 hours)
- **International School and Conference on Network Science (NetSciX 2024)**
A major international conference focusing on network science across disciplines such as physics, biology, and social systems.
My contribution: **Regular talk**
- **International School and Conference on Network Science (NetSci 2023)**
The flagship annual event for network scientists worldwide.
My contribution: **Regular talk.**

Poster Presentations

- **Statistical Physics of Cognition**
Institute of Physics, London (November 2024)
This event brings together physicists and cognitive scientists to explore the role of statistical physics in understanding cognition. My contribution: poster presentation.
- **BrainModes 2024**
Bilbao, Spain (November 2024)
A highly specialized conference focused on brain dynamics and modes of neural activity. My contribution: poster presentation.
- **Association for the Scientific Study of Consciousness (ASSC27, 2024)**
Tokyo, Japan (2024)
This leading conference focuses on advancing the scientific understanding of consciousness. Contributed by creating a poster, presented on-site by co-authors.
- **Organization for Human Brain Mapping (OHBM 2024)**
Seoul, South Korea (2024)
Designed a poster that was presented on-site by co-authors. [Abstract link](#)
- **Computational and Systems Neuroscience (Cosyne 2024)**
Lisbon, Portugal (February 2024)
A premier conference for computational and systems neuroscientists, known for its high selectivity and impact. Presented a poster.
- **NeuroDays 2023**
Marseille, France (November 2023)
An interdisciplinary neuroscience symposium showcasing regional and international talent. Presented a poster, awarded **2nd best poster**.
- **Interdisciplinary College (IK) 2023**
Marseille, France (November 2023)
A unique conference blending neuroscience, psychology, and computer science, emphasizing interdisciplinary education and collaboration. Presented a poster.

Extracurricular

- **CAM – Alba (CN), Italy – Sep 2015 - Jun 2017.**

I conducted didactic activities as a volunteer.
- **Bassist and rapper – 2012 - present**

I started to play classical guitar in 2008. A few years later, I created with some friends a jazz band, in which I play the **bass** and **sing**. Recently, I also started writing and publishing **rap** songs, following the vibes of indie and jazz.